ENTERPRISE ARCHITECTURE SPECIFICATION

Staff Management Portal — Technical Specification

Comprehensive system architecture, production deployment reference, and RESTful API specifications for the Synchronoss enterprise staff management system.

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Live Frontend UI:	**https://staff-btv0.onrender.com**	**Live Backend API:**	**https://staff-backend-mwuh.onrender.com**
Interactive Swagger UI:	https://staff-backend-mwuh.onrender.com/swagger-ui/index.html	**OpenAPI 3.0 JSON:**	https://staff-backend-mwuh.onrender.com/v3/api-docs
Health Endpoint:	https://staff-backend-mwuh.onrender.com/api/health (200 OK)	**Cloud Database:**	Aiven MySQL (defaultdb, SSL REQUIRED)
Deployment Branches:	main (Production), develop (Dev)	**Preserved Branch:**	main-otp (Dedicated Email OTP Flow)
Technology Stack:	Java 17, Spring Boot 3.3.5, React 18 (Vite), MySQL 8.0	**Specification Date:**	05 November 2026

1. System Overview & Core Architecture

The **Staff Management Portal** is a full-stack, enterprise-grade human capital and workforce administration platform. Designed with strict multi-tenant tenant data isolation, role-based authorization (RBAC), and automated company onboarding, the platform empowers organizations to track employee profiles, salaries, job designations, and organizational departments seamlessly.

Tier	Technology	Key Features / Responsibilities
Frontend Client	React 18, Vite 5, Axios, Lucide Icons	Single Page Application (SPA), responsive dashboard, JWT auth interceptor, automated 2s onboarding transition.
Backend API	Java 17, Spring Boot 3.3.5, Spring Security 6	RESTful microservices, stateless JWT filter, Spring Data JPA repositories, multi-stage Docker runtime.
API Documentation	SpringDoc OpenAPI 3.0, Swagger UI 2.6.0	Interactive live testing console, Bearer JWT security scheme, automated schema discovery.
Cloud Database	Aiven MySQL 8.0 Cloud Service	Dedicated managed MySQL, enforced SSL (sslMode=REQUIRED), HikariCP pool with auto schema management.
Hosting & CI/CD	Render Cloud Platform + GitHub Webhooks	Automated builds from GitHub main/develop, zero-downtime rolling deploys, HTTPS ingress on Port 443.

2. Security Architecture & Authentication Flows

The platform implements robust defense-in-depth security standards conforming to OWASP Top 10 guidelines:

• **Stateless JWT Authentication:** Secure HS256 HMAC tokens with 24-hour expiration containing user ID, email, and role claims.

• **Direct Admin Registration:** Seamless account creation without external SMTP barriers, auto-authenticating the admin with a fresh JWT token.

• **Automated Workspace Onboarding:** Displays a clean 2-second success confirmation and automatically routes the admin to the company setup form.

• **Direct Password Reset:** Instant credential recovery via email and password validation, with Google OAuth login as an integrated alternative.

• **Multi-Branch Preservation:** The full email OTP verification workflow is preserved in the dedicated main-otp branch.

3. REST API Endpoint Specifications

HTTP	Endpoint Path	Access	Description
GET	`/api/health`	Public	Health check probe returning service status UP and timestamp
GET	`/swagger-ui/index.html`	Public	Interactive OpenAPI 3.0 API documentation and live test UI
POST	`/api/auth/admin/register`	Public	Direct admin registration; returns JWT session token and user details
POST	`/api/auth/login`	Public	Standard credentials login returning Bearer JWT token
POST	`/api/auth/reset-password`	Public	Direct password reset by email or reset token with password validation
GET	`/api/auth/profile`	Admin	Fetch authenticated administrator profile details
GET	`/api/companies/my-company`	Admin	Fetch current company workspace associated with admin
POST	`/api/companies`	Admin	Create corporate organization workspace
PUT	`/api/companies`	Admin	Update organization details (name, industry, contact phone, website)
GET	`/api/employees`	Admin	List all employees within the isolated company workspace
POST	`/api/employees`	Admin	Create employee with salary, designation, email, and department
PUT	`/api/employees/{id}`	Admin	Update employee profile details
DELETE	`/api/employees/{id}`	Admin	Soft/hard delete employee record
GET	`/api/employees/departments`	Admin	List distinct departments and workforce distribution statistics

4. Production Deployment & Cloud Infrastructure

The platform is deployed in a high-availability cloud configuration:

• **Backend Web Service (Render):** Running on eclipse-temurin:17-jre-jammy packaged via multi-stage Docker build. Exposed on port 8080 (internal) with automatic HTTPS edge routing.

• **Frontend Client (Render):** Hosted as a static web application built with Vite, routing all API requests dynamically to https://staff-backend-mwuh.onrender.com.

• **Database (Aiven MySQL):** Hosted on Aiven cloud with SSL encryption (sslMode=REQUIRED), automatic failover, and HikariCP connection pooling.

• **Uptime Monitoring:** Health probe configured at https://staff-backend-mwuh.onrender.com/api/health with 5-minute polling interval to maintain active container state.